

Po-Han Chen

✉ pohan@stanford.edu | 🌐 <https://pohantw.github.io>

Education

Stanford University

Stanford, CA

Ph.D. in Electrical Engineering

2021–2026

Advisor: Priyanka Raina

Thesis Title: *Designing CGRAs for Sparse ML Acceleration and Scalable Physical Implementation*

National Tsing Hua University

Hsinchu, Taiwan

M.S. of Electrical Engineering

2016–2018

B.S. of Electrical Engineering and Computer Science

2012–2016

Experiences

SEG Pixel IP Design Team, Apple

Cupertino, CA

Hardware Design Intern

2022

- Explored and implemented hardware architectures for a newly developed internal algorithm, achieving significant area reduction compared with the legacy design.

Camera ISP Pipeline Hardware Design Team, MediaTek

Hsinchu, Taiwan

Hardware Designer

2019–2020

- Designed and verified digital IPs for camera ISP pipelines, including functional verification, coverage analysis, and synthesis.

- Developed automation tools to streamline internal hardware design and verification workflows.

Skills

HW/SW Languages: Verilog, SystemVerilog, C, Python, Tcl, Matlab

EDA Tools & Others: Simulators (VCS, Xcelium), Linting (Spyglass, nLint), Synthesis (DC, Genus),

P&R (Innovus, ICC), LEC, DRC/LVS, Verdi, Git

Selected Publications

Full list: <https://pohantw.github.io>

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, R. Lu, S. G. Ravipati, K. Koul, A. Carsello, Y. Mei, M. Horowitz, and P. Raina, “Opal: A 16-nm Coarse-Grained Reconfigurable Array SoC for Full Sparse Machine Learning Applications,” *IEEE Solid-State Circuits Letters (SSC-L)*, 2025.

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, K. Koul, S. G. Ravipati, Y. Mei, R. Lu, A. Carsello, M. Horowitz, and P. Raina, “Opal: A 16nm Coarse-Grained Reconfigurable Array for Full Sparse ML Applications,” *IEEE Custom Integrated Circuits Conference (CICC)*, 2025.

P.-H. Chen, C. Tsao, and P. Raina, “An Open-Source 4×8 Coarse-Grained Reconfigurable Array Using SkyWater 130 nm Technology and Agile Hardware Design Flow,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023.

P.-H. Chen, S.-W. Yang, and C.-T. Huang, “A 250-mW 5.4G-Rendered-Pixel/s Realistic Refocusing Processor for High-Performance Five-Camera Mobile Devices,” *IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS)*, 2023.

P.-H. Chen, S.-W. Yang, S.-Y. Huang, L.-D. Chen, and C.-T. Huang, “A 250mW 5.4G-Novel-Pixel/s Photorealistic Refocusing Processor for Full-HD Five-Camera Applications,” *IEEE Symposium on VLSI Circuits (VLSI)*, 2019.